

BLUESTOCK MUTUAL FUND ANALYTICS PROJECT

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Institute : Thapar Institute of Engineering & Technology

Project : Mutual Fund Analytics & Investment Insights

Github : https://github.com/Sumit21Sinha/BlueStocks_RealProject



Tools & Technologies

- Python
- Pandas
- NumPy
- SQLite
- SQL
- Matplotlib
- Seaborn
- Power BI
- Git & GitHub

Executive Summary

The objective of this project was to build a complete Mutual Fund Analytics platform capable of ingesting, cleaning, analyzing, and visualizing mutual fund data. The project focuses on transforming raw mutual fund datasets into actionable business insights through data engineering, exploratory data analysis, advanced financial analytics, and interactive dashboarding.

The solution was developed using Python, SQL, and Power BI. Historical NAV data, mutual fund performance metrics, investor transaction records, portfolio holdings, SIP statistics, folio growth data, and AUM trends were integrated into a unified analytics workflow.

The project provides fund ranking mechanisms, risk-return analysis, investor behavior insights, SIP growth trends, sector concentration analysis, and risk metrics such as Sharpe Ratio, Maximum Drawdown, Value-at-Risk (VaR), and Conditional Value-at-Risk (CVaR).

The final outcome is an end-to-end analytics system that enables investors and analysts to evaluate mutual fund performance and make data-driven investment decisions.

1. Introduction

Mutual funds have become one of the most popular investment vehicles due to professional management, diversification benefits, and accessibility for retail investors.

However, selecting the right fund requires analyzing multiple dimensions including:

- Historical performance
- Risk-adjusted returns
- Portfolio diversification
- Sector exposure
- Investor participation trends
- Market volatility

This project addresses these challenges by creating a centralized analytics platform that evaluates mutual funds using multiple quantitative metrics.

2. Project Objectives

The major objectives of the project are:

1. Build an end-to-end data analytics pipeline.
2. Clean and preprocess raw mutual fund datasets.

3. Store processed data in a structured database.
4. Perform exploratory data analysis.
5. Calculate advanced financial metrics.
6. Rank mutual funds using a composite scoring framework.
7. Analyze investor behavior and industry trends.
8. Build interactive Power BI dashboards.
9. Generate actionable investment insights.

3. Dataset Description

The project uses multiple datasets.

3.1 Mutual Fund Performance Dataset

Contains:

- Fund Name
- Fund House
- CAGR Returns
- Alpha
- Beta
- Sharpe Ratio
- Sortino Ratio
- Expense Ratio
- Risk Grade
- AUM

Purpose:

Evaluate fund performance and risk.

3.2 NAV History Dataset

Contains:

- AMFI Code
- Date

- NAV

Purpose:

Calculate daily returns, volatility, CAGR, VaR, and CVaR.

3.3 Investor Transactions Dataset

Contains:

- Investor ID
- Transaction Date
- Transaction Type
- Amount
- City Tier
- Income Group
- Gender
- Age Group

Purpose:

Analyze investor behavior and transaction patterns.

3.4 SIP Industry Dataset

Contains:

- Monthly SIP Inflows
- Active SIP Accounts
- SIP AUM

Purpose:

Understand industry growth trends.

3.5 Folio Dataset

Contains:

- Total Folios
- Equity Folios
- Debt Folios

- Hybrid Folios

Purpose:

Track investor participation growth.

3.6 Portfolio Holdings Dataset

Contains:

- Stock Name
- Sector
- Weight Allocation

Purpose:

Measure portfolio diversification and concentration risk.

4. ETL & Data Cleaning Process

The ETL pipeline was developed using Python.

Data Validation Performed

NAV Dataset

- Missing value check
- Date conversion
- Duplicate removal
- Invalid NAV detection
- Forward filling missing values

Transactions Dataset

- Duplicate removal
- Invalid transaction detection
- Date standardization
- KYC validation

Scheme Performance Dataset

- Missing value analysis
- Negative Sharpe detection
- Expense ratio validation

Result:

All datasets were cleaned and standardized before loading into SQLite.

5. Database Design

SQLite was used as the project database.

Tables Created

Dimension Tables

- dim_fund

Fact Tables

- fact_nav
- fact_transactions
- fact_performance

Advantages:

- Fast querying
- Structured storage
- Easy Power BI integration

6. SQL Analysis

Several analytical SQL queries were developed.

Examples:

- Top funds by Sharpe Ratio
- Top funds by CAGR
- Highest AUM fund houses
- Investor distribution analysis
- Risk category analysis
- Fund ranking reports

SQL enabled rapid aggregation and business intelligence reporting.

7. Exploratory Data Analysis

EDA was performed using Matplotlib and Seaborn.

Visualizations Created

Fund Analysis

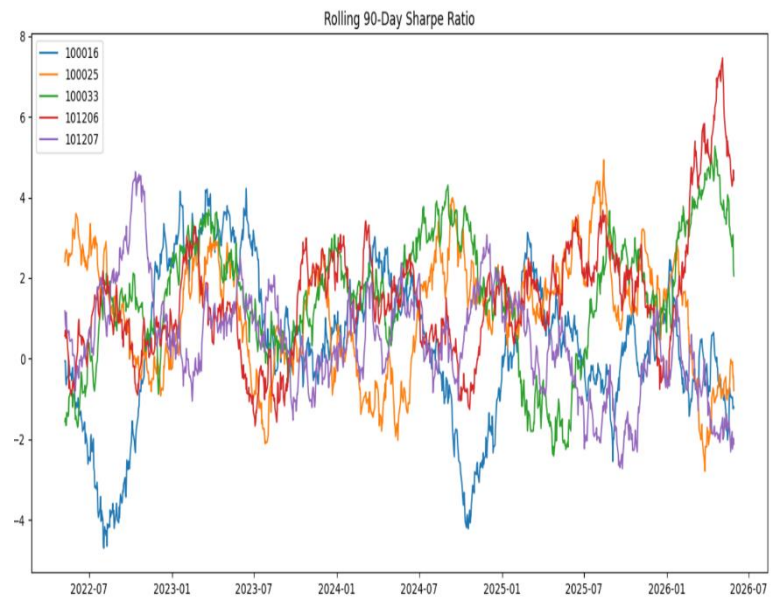
- Sharpe Ratio Distribution
- CAGR Distribution
- Risk Category Analysis
- Expense Ratio Analysis

Industry Analysis

- SIP Growth Trend
- Folio Growth Trend
- AUM Growth Trend

Portfolio Analysis

- Sector Allocation Pie Chart
- Fund Correlation Heatmap



8. Advanced Analytics

8.1 Daily Returns

Daily returns were calculated using:

$$\text{Return} = (\text{Current NAV} - \text{Previous NAV}) / \text{Previous NAV}$$

Purpose:

Measure day-to-day fund performance.

8.2 CAGR

Compound Annual Growth Rate:

Used to evaluate long-term fund performance.

8.3 Volatility

Standard deviation of returns.

Purpose:

Measure investment risk.

8.4 Sharpe Ratio

Sharpe Ratio = (Return – Risk Free Rate) / Volatility

Purpose:

Evaluate risk-adjusted performance.

8.5 Maximum Drawdown

Measures largest decline from peak to trough.

Purpose:

Quantify downside risk.

8.6 Value at Risk (VaR)

95% confidence loss estimate.

Purpose:

Estimate potential losses under normal market conditions.

8.7 Conditional Value at Risk (CVaR)

Average loss beyond VaR threshold.

Purpose:

Measure extreme downside exposure.

9. Mutual Fund Ranking Model

A composite scoring framework was created.

Factors:

- CAGR
- Sharpe Ratio
- Maximum Drawdown

Final Score:

Weighted combination of all metrics.

Benefits:

- Balanced evaluation
- Reduced single-metric bias
- Better investment recommendations

10. Key Findings & Insights

Finding 1

Monthly SIP inflows increased significantly from approximately ₹11,000 Crore to over ₹30,000 Crore.

Insight

Retail participation in mutual funds is growing rapidly.

Finding 2

Total folios increased from nearly 13 Crore to over 26 Crore.

Insight

The mutual fund industry continues to expand.

Finding 3

A small group of funds consistently outperformed peers.

Insight

Performance is highly concentrated among top-ranked schemes.

Finding 4

Higher returns are generally associated with higher volatility.

Insight

Risk-return tradeoff remains evident across funds.

Finding 5

Top-ranked funds maintained superior Sharpe Ratios.

Insight

Risk-adjusted returns provide better investment guidance than raw returns alone.

Finding 6

T30 cities contribute the majority of investments.

Insight

B30 markets remain a major growth opportunity.

Finding 7

Several portfolios exhibited significant sector concentration.

Insight

Diversification remains critical for risk management.

Finding 8

VaR and CVaR analysis identified funds with elevated downside exposure.

Insight

Investors should evaluate downside risk before investing.

11. Dashboard Development

Power BI dashboards were developed.

Page 1 – Executive Overview

KPIs:

- Total Funds
- Average CAGR
- Average Sharpe Ratio
- Average Fund Score

Visuals:

- Top Ranked Funds
- Top CAGR Funds

Page 2 – Fund Performance

Visuals:

- Top Sharpe Funds
- Most Volatile Funds
- Lowest Drawdown Funds
- Risk vs Return Scatter Plot

Page 3 – Investor Insights

Visuals:

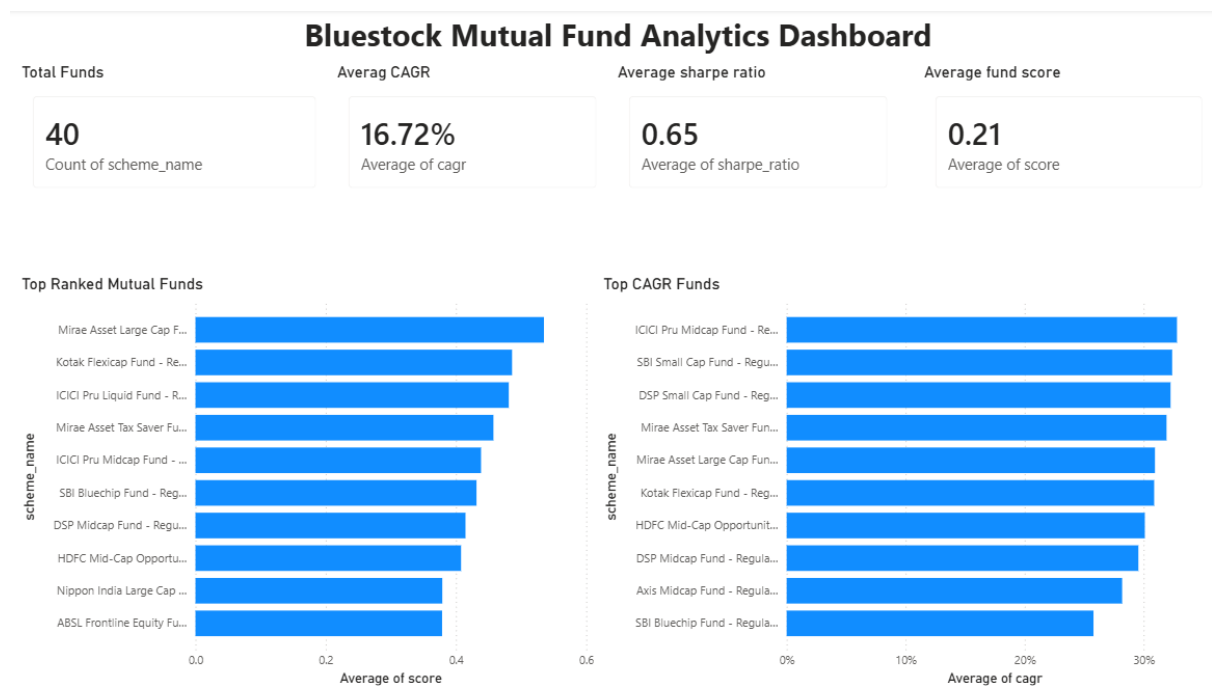
- Investor Distribution
- City Tier Analysis
- Income Segment Analysis
- Gender Analysis

Page 4 – Industry Trends

Visuals:

- SIP Growth Trend
- Folio Growth Trend
- Fund House Comparison

12. Dashboard Screenshots



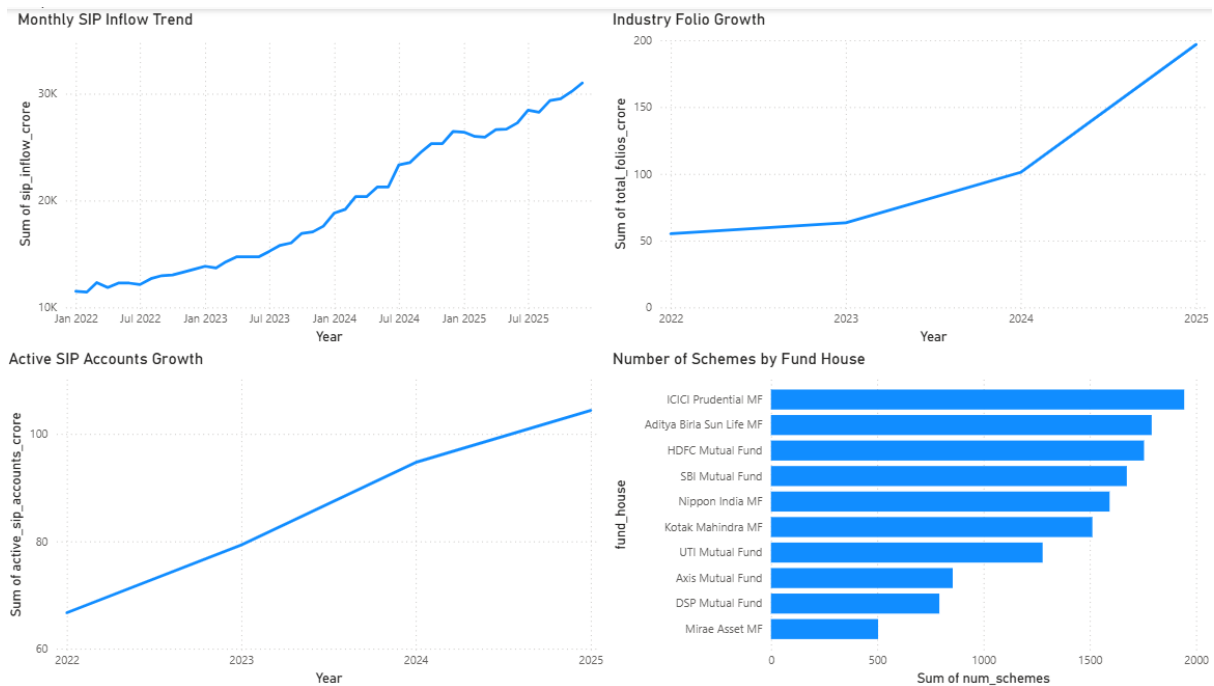
Screenshot 1: Executive Overview Dashboard



Screenshot 2: Fund Performance Dashboard



Screenshot 3: Investor Insights Dashboard



Screenshot 4: Industry Trends Dashboard

13. Recommendations

1. Prioritize funds with consistently high Sharpe Ratios.
2. Use risk-adjusted performance metrics rather than returns alone.
3. Monitor VaR and Drawdown before investment decisions.

4. Increase focus on B30 market expansion.
5. Promote SIP investments due to strong long-term growth.
6. Avoid excessive sector concentration.

14. Limitations

- Historical performance does not guarantee future returns.
- Limited mutual fund universe.
- Risk-free rate assumptions may vary.
- No real-time market integration.

15. Future Scope

Future enhancements include:

- Machine Learning fund recommendation system.
- Real-time NAV API integration.
- Portfolio optimization engine.
- Streamlit deployment.
- Cloud-hosted analytics platform.
- Automated investor segmentation.

Conclusion

This project successfully developed a complete Mutual Fund Analytics platform integrating data engineering, financial analytics, SQL querying, risk modeling, and dashboard visualization. The solution demonstrates how data analytics can be leveraged to transform raw mutual fund information into meaningful investment insights. The project provides a strong foundation for advanced financial analytics applications and real-world investment decision support systems.

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